

Answers to the question no. (1).

(a) Ethics: Refers to fundamental moral principles governing right and wrong actions beyond written rules. In the health app scenario, ethics demands respecting users autonomy and privacy, leading developers to reject using health and location data for targeted ads without explicit consent.

Law: Enforceable rules established by governmental authority. Legally, if current privacy statutes or user agreements contain loop-hole or lack explicit restrictions on data reuse, the company might proceed without immediately breaking any laws.

Professional Responsibility: Standard of Conduct expected by professional bodies (such as ACM or IEEE). Computing professionals are obligated to protect users well-being and privacy regardless of legal loopholes or marketing pressure, directing the team to refuse unauthorized data exploitation.

(b) 1. Therac-25 Radiation Therapy Machine

Technology involved: Software-controlled medical linear accelerators.

Ethical Failure: Negligent software engineering; lack of defensive programming, removal of hardware safety interlocks and failure to investigate early bug reports.

Major lesson learned: Software quality in safety-critical systems is a direct safety issue; hardware fail-safes and rigorous testing are mandatory.

2. Volkswagen Emissions Deception (1)

Technology involved: Engine control unit firmware

Ethical Failure: Software engineers intentionally programmed defeat devices to detect test environments and lower emissions artificially.

Answer to the question no. (2)

(a) Utilitarianism vs Deontology in AI Data Collection:

Utilitarianism: Evaluates actions by overall net outcome ($\text{Utility} = \text{Benefits} - \text{Harms}$). It balances engagement gains for millions against privacy intrusion harms.

Deontology: Focuses on absolute moral duties and right. It treats users as ends rather than means, deeming private message scanning without explicit consent an absolute violation of moral duty, regardless of engagement benefits.

(b) Virtue Ethics in Software Engineering Decisions:

Concept: Emphasizes moral character traits (virtues) over rigid rules or calculations.

Honesty and Respect: Requires transparent disclosure of data collection practices to users.

Fairness and Responsibility: Ensures algorithms do not exploit vulnerable user networks.

Courage: Empowers engineers to challenge management when privacy boundaries are crossed.

Answers to the question no. (3)

(a) Importance of Transparency and Explainability:

Human agency: Allows individuals to understand automated decisions affecting their livelihood.

Bias Detection: Enables auditors to identify systemic discrimination in credit, hiring or medical algorithms.

Due Process: Guarantees that decisions rely on accurate data rather than invalid correlations.

(b) Societal Harms of Black-Box AI:

Systemic Exclusion: Perpetuates historical socio-economic discrimination without opportunity for audit or correction.

Erosion of Public Trust: Reduces public confidence in financial and civic institutions when automated denials cannot be justified.

Answers to the question no. (4)

(a) Key Principles and Importance of ACM code:

Key Principles: "Contribute to human well-being", "Avoid Harm", "Be honest and trustworthy" and "Respect Privacy".

Importance: Provides an unified ethical benchmarks, safeguards public interests and offers engineers professional grounds to resist unsafe organizational directives.

(b) Application to Forced Premature Release Scenario:

ACM 1.2 (Avoid Harm): Deploying code with known security vulnerabilities risks system compromise and users harm.

ACM 2.5 (High Quality): Developers must achieve high quality in process and work products.

Action: The engineers must refuse deployment document security risks and escalate concerns to management.

Answers to the question no. (b)

(a) Professional Dissent vs Whistleblowing

Professional Dissent: Internal disagreement expressed within organizational channels regarding technical, security or ethical concerns.

Whistleblowing: Disclosing internal misconduct or safety risks to external authorities or the public when internal channels fail.

(b) Ethical Considerations Before External Disclosure:

Severity of Harm: Assessing if there is an imminent, severe threat to public safety or privacy.

Verification of Facts: Ensuring evidence is documented, accurate and technically verified.

Exhaustion of Internal Remedies: Confirming Management was formally informed and given an opportunity to remediate.

(b) Whistleblowing: Disclosing information to external authorities or the public when internal channels fail.

Answers to the question no. (6)

(a) Conflict Between Well-Being and Privacy:

Contribute to Well-Being: Analyzing location data helps track disease spread and save life.

Respect Privacy: Tracking mobile locations exposes personal habits, residences and daily routines.

Conflict: Maximizing public health tracking can erode individual privacy rights.

(b) Guiding Principles for Personal Data Collection:

Informed Consent: Obtaining clear user permission.

Purpose Limitation: Using data strictly for public health not commercial analysis.

Data Minimization: Collecting only essential points.

Security and Transparency: Encrypting datasets, and making data policies public.

Answer to the question no. (7)

(a) Distinctions in IP Protection:

Copyright: Protects original literary/artistic expressions, including actual source code.

Patent: Protects novel, non-obvious, useful technical processes or functional algorithms.

Trademark: Protects brand identity markers such as product names, logos and slogans.

(b) Asset Protection Mapping:

Source code → Copyright: Prevents unauthorized copying of exact code structures.

Hardware-Assisted Algorithm → Patent: Protects the functional technical process.

Product Name & Logo → Trademark: Safeguards brand identity and prevents market confusion.

Answers to the question no. (8)

(a) Comparison of Licensing Models:

Feature	Open-Source Licensing	Proprietary Licensing
Source code Access	Publicly accessible for inspection and altering	Closed source
Commercial Control	Limited direct control emphasizes wide adoption	High commercial control and direct monetization
Modification Rights	Granted under license terms (e.g; GPL, MIT).	Prohibited without specific commercial agreement

(b) Ethical Advantages and Concerns:

Open-Source: Advantage: Community collaboration and security transparency. Concern: Potential exploitation without contributing fixes back.

Proprietary: Advantage: Sustainable funding for research and development. Concern: Vendor lock-in and non-auditable black-box code.

Answer to the question no. (9):

(a) Responsible Disclosure Defined:

Responsible Disclosure: Reporting security vulnerabilities privately to vendors and giving them time to patch before public release.

Cybercrime/Exploitation: Secretly exploiting vulnerabilities for personal gain, theft or unauthorized entry.

(b) Ethical Guidelines for Researchers:

Avoid Unnecessary Access: Do not view or exfiltrate actual users data.

Provide Proof-of Concept: Deliver minimum technical details necessary for vendors reproduction.

Maintain Confidentiality: Keep vulnerability details secure during remediation.

Answers to the question no. (10)

(a) Ethical Hacking vs Cybercrime

Ethical Hacking: Authorized security evaluation performed within agreed limits to strengthen systems.

Cybercrime: Unauthorized access, service disruption or data theft conducted illegally.

(b) Conditions of Ethical Hacking:

- i) Explicit Authorization: Written authorization from asset owners.
- ii) Defined Scope and Timing: Adhering strictly to permitted systems and operational hours.
- iii) Proportionality & Non-Disruption: Ensuring tests do not disrupt real services or breach data privacy.

Answers to the question no. (11)

(a) Definition and Causes of Algorithmic Bias:

Definition: Systemic, unfair discrimination in automated outputs impacting specific demographic groups.

Causes:

- i) Biased Training Data

- ii) Proxy Variables

- iii) Sampling Imbalance

- iv) Inappropriate objectives

(b) Detection & Measurement

Disparate Impact Analysis: Evaluating selection ratios across demographic groups.

Fairness Metrics Auditing: Checking equalized odds and equal opportunity scores during validation.

Answers to the question no. (12)

(a) Concept of Accountability: Autonomous AI systems operate independently, but moral and legal accountability remains strictly with human creators, deployers and organizations.

(b) Responsibility Allocation:

Developers/Testers: Responsible for software quality boundary-case testing and code safety.

Manager's/Manufacturers: Responsible for setting safety standards, dataset choices and operational oversight.

Operators: Responsible for deployment adherence and operational monitoring.

Answers to the question no. (13)

(a) Ethical Leadership: Organizational leadership prioritizing integrity, safety and accountability over delivery speed, cultivating a culture where safety concerns are encouraged.

(b) Establishing an Ethical Culture:

Clear Policies and Training: Regular ethics education and explicit safety protocols.

Aligned Incentives: Rewarding security compliance alongside production velocity.

Open channels: Maintaining transparent reporting mechanisms and whistleblowers protections.

Answer to the question no. (14)

(a) Informed Consent: Voluntary, clear and uncoerced permission given by a fully informed user regarding data processing activities.

(b) Lengthy Policies vs. Meaningful Consent:

Requiring users to agree to lengthy legal documents does not constitute meaningful consent because users rarely comprehend dense legal jargon under "click-wrap" conditions.

Answers to the question no. (15)

(a) Major Ethical Concerns:

Loss of Anonymity: Creates a chilling effect on public assembly and free expression.

Inaccuracy and Missidentification: Algorithmic errors disproportionately impacting specific demographic groups.

Function Creep: Unregulated repurposing of monitoring systems beyond original scope.

(b) Facial recognition Benefits vs. Risks:

Benefits: Rapid suspect identification and public safety enhancement.

Risks: Warrantless tracking, privacy loss and potential false arrests.

Answers to the question no. (16)

(a) Copyright vs. Patents:

Copyright: Automatically protects original written code expressions and graphical interface designs.

Patents: Protects novel, non-obvious technical processes, functionality or algorithms through formal registration.

(b) Asset Mapping:

Source Code & GUI Layout: Protected under copyright

Novel Compression Algorithm: Potentially eligible for patent protection.

Answer to the question no. (17)

(a) Digital Divide: The inequality between demographic groups in access to, use of, or knowledge of information and communication technologies.

(b) Social Implications:

Education & Employment: Limits access to online learning and remote job opportunities

Essential Services: Restricts access to modern healthcare services and government assistance.

Answers to the question no. (18)

(a) Green Computing: The environmentally sustainable design, production, operation and disposal of computing resources.

(b) Strategies for Reducing Environmental Impact:

Resource Optimization: Cloud virtualization and server workload balancing.

Lifecycle Care: E-waste recycling, hardware refurbishment, and energy-efficient software design.

Answer to the question no. (19)

(a) Conflict of Interest: A situation where personal or family relationships compromise or appears to compromise professional objectivity and judgement.

(b) Significance of Conflict Types:

Actual: Directly compromises purchasing or design decisions.

Potential: Risk of future compromise as vendor evaluation progresses.

Perceived: Undermines public trust and team credibility regardless of actual bias.

Answers to the question no. (20)

(a) Core Issues Identified:

Cybersecurity Risk: Unpatched security flaws permitting unauthorized data access.

Privacy Concern: Potential exposure of private household behaviour.

Professional conflict: Release schedule pressure versus professional duty of safety.

(b) Framework Evaluations:

Utilitarianism: Premature launch risks security breaches, causing far greater net harm than financial delays.

Virtue Ethics: Violates professional standards of honesty, integrity and responsibility.

Deontology: Launching unsafe products
breaches the fundamental obligation to
safeguard users.

Cybersecurity Risk: Unpatched security flaws
permitting unauthorized data access.

Privacy Concern: Potential exposure of private
household behaviors.

Professional conflict: Release schedule pressure
versus professional duty of safety.

(b) Framework Evaluations:

Utilitarianism: Promotes launch risks, security
breaches, causing two greatest net human benefit
financial delays.

Viola Ethics: Violates professional standards
of honesty, integrity and responsibility.